

EMERSON EIMS

BUILDING SUITE PRO

Full QS Test

Location: Nakuru, Kenya
Total Area: 220 m² | Units: 1 | Storeys: 2
Architectural Style: Modern
Date: 27 April 2026

PROJECT INFORMATION	
Project ID	73515517-1001-47bc-b210-c1bc82a72904
Location	Nakuru, Kenya
Total Area	220 m²
Units / Apartments	1
Storeys	2
Bedrooms per Unit	4
Foundation Type	Strip + Ground Beam
Style	Modern
Report Generated	2026-04-27 04:51

TABLE OF CONTENTS

1. Validation Engine — Audit Trail
2. Design Brief & Design Criteria (incl. EC1 design loads)
3. Structural Design Calculations (Slab · Beam · Column · Foundation)
4. Bar Bending Schedule (BS 8666)
5. MEP Design — Electrical, Plumbing & Drainage
6. Bill of Quantities (KQS-2025)
7. Statutory Compliance Pack & Professional Certification
- 7B. QS Quotation — Priced Offer (BOQ · Cost Summary · Payment Schedule · T&Cs;)
8. Project Summary & Phase Data
9. Site Plan
10. Architectural Floor Plan
11. Building Elevations (North & East)
12. Building Cross Section
13. Structural Layout — Columns, Beams & Foundation
14. MEP Layout — Electrical
15. MEP Layout — Plumbing
16. Cost Summary

VALIDATION ENGINE — AUDIT TRAIL

■ 1 BLOCKER(S) — DO NOT SUBMIT

• Kenya statutory IDs missing: Plot / cadastral reference; Architect (must be registered with BORAQS); Structural Engineer (must be registered with EBK); Quantity Surveyor (must be registered with BORAQS). Required for permit submission to County Government Building Plans Section.

1 auto-correction(s) applied with code citation:

Field	Original	Auto-corrected to	Reason / Code citation
Foundation	strip	strip + ground beam	2-storey → strip + ground beam (BS 8004 §7)

Compliance Status: BLOCKED — DO NOT SUBMIT

1. DESIGN BRIEF & DESIGN CRITERIA

Item	Value
Project name	Full QS Test
Location	Nakuru, Kenya
Total floor area	220.0 m²
No. of storeys / floor count	2 (single source of truth)
Dwelling units	1
Bedrooms / unit	4
Architectural style	Modern
Foundation	STRIP + GROUND BEAM
Currency	KES
Plot / LR No.	— REQUIRED —
Coordinates (lat,lng)	-1.2921,36.8219
Title Deed	—
Architect (BORAQS)	— REQUIRED —
Structural Engineer (EBK)	— REQUIRED —
Quantity Surveyor (BORAQS)	— REQUIRED —
Client	Smoke Test

Design loads (BS EN 1991-1-1 Cat A — residential):

Action	Value (kN/m²)	Reference
Permanent action g_k (slab + finishes + partitions)	7.00	EC1 Table 6.1
Variable action q_k	2.00	EC1 Table 6.2 Cat A
Design ULS $w = 1.35 g_k + 1.5 q_k$	12.45	EC0 §6.4.3.2
Design SLS $w = g_k + q_k$	9.00	EC0 §6.5.3

2. STRUCTURAL DESIGN CALCULATIONS

Calculations to BS EN 1992-1-1 (EC2) using $f_{ck} = 30 \text{ MPa}$, $f_{yk} = 500 \text{ MPa}$, cover per exposure class XC1 internal / XC3 external. All members hand-verified against simplified Eurocode 2 design rules; final detailing to be reviewed by the Structural Engineer of Record (EBK-registered) before issue for construction.

2.1 Slab design (one/two-way RC slab)

Parameter	Value
Short span L_x	4.50 m
Long span L_y	4.50 m
Behaviour	Two-way
Slab thickness h	175 mm (span/depth = 26)
Effective depth d	144.0 mm
Cover	25 mm (XC1)
Design moment M	15.63 kN·m / m
As required	263.0 mm ² / m
As minimum (EC2 §9.3.1.1)	228.0 mm ² / m
As provided	263.0 mm ² / m
Reinforcement	T12 @ 300 c/c BW

2.2 Beam design (simply-supported RC beam)

Parameter	Value
Clear span L	4.50 m
Beam section $b \times h$	225 × 375 mm
Effective depth d	329.0 mm
Cover	30 mm
Service load w	58.13 kN/m
Design moment M	147.2 kN·m
Design shear V	130.8 kN
As required (tension)	1082.0 mm ²
Main steel	6 no. T16 bot · 3 no. T16 top
Shear links	T8 @ 225 c/c

2.3 Column design (axially loaded RC column)

Parameter	Value
Axial load N (ULS)	514.0 kN
Section	225 × 225 mm
Concrete grade	C25/30 ($f_{ck} = 30 \text{ MPa}$)
Steel grade	$f_{yk} = 500 \text{ MPa}$
Steel ratio ρ	0.020 (2.0 %)
As required	1012.0 mm ²
Main steel	6 no. T16
Links	T8 @ 192 c/c

2.4 Foundation design (per BS 8004 / EN 1997-1)

Parameter	Value
Foundation type	STRIP + GROUND BEAM
Plan size	B = 590 mm wide × Df = 1200 mm deep
Thickness	300 mm
Concrete grade	C25/30
Cover	75 mm (XC2 ground contact)
Safe bearing pressure	220.0 kPa (lab-tested — geotech report NAK-GEO-2026-0117)
Soil classification (USCS)	CL
Water-table depth below GL	4.2 m (lab-measured)
Geotechnical Engineer	Eng. Kariuki
Column load to foundation	514.0 kN
Reinforcement	T12 @ 150 c/c BW bottom + T10 @ 200 distribution top

✓ Foundation parameters taken from on-site geotechnical investigation (report ref. NAK-GEO-2026-0117, signed by Eng. Kariuki).
Compliant with BS EN 1997-1 / NCC 2.2.5.

3. BAR BENDING SCHEDULE (BS 8666)

Schedule of all reinforcement bars with mark, diameter, shape code, length per bar, number off, total length and weight. Steel grade 500B. Bend radii and bar shapes comply with BS 8666:2020 Table 2. Quantities to be verified by the Engineer before cutting and bending.

Bar Mark	Dia.	Shape Code	Length per bar (m)	No. off	Total length (m)	Weight (kg)
S1	T12	00	15.13	88	1331.65	1182.5
B1-bot	T16	00	5.00	170	850.00	1342.1
B1-top	T16	37	5.00	68	340.00	536.9
B1-link	T8	51	0.85	680	578.00	228.3
C1	T16	00	3.50	72	252.00	397.9
C1-link	T8	51	0.60	216	129.60	51.2
F1	T12	00	3.00	144	432.00	383.6
					TOTAL	4122.5

Total reinforcement steel: 4,122 kg $\approx$ 4.12 tonnes

4. MEP DESIGN — ELECTRICAL, PLUMBING & DRAINAGE

Electrical design per IEC 60364 / KS IEC 60364. Plumbing & sanitary drainage per BS EN 12056-2 and Kenya NCC Building Code SEHS. Septic tank to NEMA EIA Regulations 2003.

4.1 Electrical load schedule

Load description	Value
Lighting (LED) @ 5 W/m ²	1100 W
Small power sockets @ 25 W/m ²	5500 W
Cooker / hob (per unit)	6000 W
Water heaters	6000 W
Air conditioning (split units)	6000 W
TOTAL CONNECTED LOAD	24,600 W
Diversity factor	0.65
DESIGN MAXIMUM DEMAND	15,990 W (16.8 kVA)
Main breaker (single phase 230 V)	75 A

4.2 Final-circuit schedule

Circuit	Load	Cable	Protection	RCD
L1-Ln (Lighting)	1100 W	1.5 mm ² T+E	10 A B-curve MCB	—
P1-Pn (Sockets)	5500 W	2.5 mm ² T+E	20 A B-curve MCB	30 mA RCD
C (Cooker)	6000 W	6 mm ² T+E	32 A B-curve MCB	—
W (2× water heater)	6000 W	4 mm ² T+E	25 A B-curve MCB	30 mA RCD
A1-A4 (AC splits)	6000 W	2.5 mm ² T+E	16 A C-curve MCB	—

4.3 Single-line diagram (text representation)

KPLC supply ■ kWh meter ■ Main isolator 75 A ■ Main DB (Type-B SPD)
■ Lighting circuits L1...Ln (1.5 mm² · 10 A)
■ Socket circuits P1...Pn (2.5 mm² · 20 A · 30 mA RCD)
■ Cooker C (6 mm² · 32 A)
■ Water heaters W (4 mm² · 25 A · 30 mA RCD)
■ AC splits A1...An (2.5 mm² · 16 A)
■ Earth busbar ■ copper earth-rod ≤ 1 Ω

4.4 Plumbing fixture schedule

Fixture	Qty
Water Closet (WC)	2
Wash-Hand Basin	2
Shower / Bath	2
Kitchen Sink	1
Laundry Sink	1
External Bib Tap	2

4.5 Drainage gradient table (BS EN 12056-2)

Pipe	Slope range	Min fall
100 mm soil pipe (WC)	1:40 to 1:80	≥ 1.00 m fall per 1 m
75 mm waste	1:40	0.025 m / m

50 mm waste	1:40	0.025 m / m
32 mm WHB / sink	1:20	0.050 m / m

4.6 Septic tank & soak-pit sizing

Parameter	Value
Design occupancy	8 persons (BS 8233 — bedrooms × 2)
Daily wastewater flow (180 L/p/d)	1,600 L/day
Septic tank volume (3-day retention)	4.8 m ³
Suggested tank dimensions	2.5 × 1.2 × 1.6 m (L×W×D)
Soak-pit area (porous soil, 30 L/m ² /d)	53.33 m ²
Soak-pit diameter (circular)	8.24 m

■ Final septic-system geometry must be confirmed by an on-site percolation test. A NEMA Environmental Impact Assessment (EIA) project report is required for systems > 50 persons or sites near surface watercourses.

5. BILL OF QUANTITIES (KQS-2025)

Contractor-grade BOQ in KES, structured by trade per the Kenya Quantity Surveyors Standard Method of Measurement. Rates are KQS-2025 indicative urban rates; tenderers shall price all items. All quantities to be verified on site.

QS rate edits applied: 1 line item(s) priced at QS-overridden rates (marked ★ in the Rate column). Auto-calculated rates shown in parentheses for audit trail.

A — PRELIMINARIES

Description	Unit	Qty	Rate	Amount
Site mobilisation, hoarding, signage	item	1.00	KSh 180,000	KSh 180,000
Insurance (CAR + WCA + Public Liability)	item	1.00	KSh 320,000	KSh 320,000
NCA Levy (0.5 % of works)	item	1.00	KSh 95,000	KSh 95,000
Site office, store, sanitation, water & power	item	1.00	KSh 165,000	KSh 165,000
Site supervision (Resident Engineer / Clerk of Works)	months	8.00	KSh 85,000	KSh 680,000
			Trade subtotal	KSh 1,440,000

B — SUBSTRUCTURE

Description	Unit	Qty	Rate	Amount
Site clearance & strip topsoil	m²	286.00	KSh 120	KSh 34,320
Bulk excavation foundation trenches	m³	66.00	KSh 650	KSh 42,900
Hardcore filling 200 mm	m³	228.00	★ KSh 12,345 (auto KSh 850)	KSh 2,715,900
Lean concrete blinding 50 mm	m³	11.00	KSh 14,000	KSh 154,000
RC foundation C25/30	m³	66.00	KSh 26,000	KSh 1,716,000
			Trade subtotal	KSh 4,663,120

C — SUPERSTRUCTURE

Description	Unit	Qty	Rate	Amount
RC columns C25/30	m³	4.25	KSh 32,500	KSh 138,206
RC beams C25/30	m³	25.82	KSh 30,500	KSh 787,472
RC suspended slabs C25/30	m³	77.00	KSh 28,500	KSh 2,194,500
Reinforcement bars (BBS)	kg	4,122.50	KSh 165	KSh 680,212
Formwork to soffits & sides	m²	642.43	KSh 1,450	KSh 931,520
200 mm machine-cut block walling	m²	355.98	KSh 1,800	KSh 640,760
Plaster both sides	m²	711.96	KSh 1,100	KSh 783,151
			Trade subtotal	KSh 6,155,821

D — ROOFING

Description	Unit	Qty	Rate	Amount
Timber roof trusses & purlins	m²	220.00	KSh 2,200	KSh 484,000
0.6 mm box-profile coloured iron sheet	m²	231.00	KSh 1,450	KSh 334,950
9 mm gypsum ceiling on softwood brandering	m²	440.00	KSh 1,850	KSh 814,000
			Trade subtotal	KSh 1,632,950

E — FINISHES

Description	Unit	Qty	Rate	Amount
Floor ceramic / porcelain tiling	m²	374.00	KSh 2,400	KSh 897,600

Wall tiling (wet areas)	m ²	79.20	KSh 2,200	KSh 174,240
Internal paint (Crown / Plascon emulsion)	m ²	569.56	KSh 520	KSh 296,173
External paint (weather-shield)	m ²	284.78	KSh 720	KSh 205,043
			Trade subtotal	KSh 1,573,056

F — JOINERY

Description	Unit	Qty	Rate	Amount
Internal flush doors with hardware	each	7.00	KSh 18,000	KSh 126,000
External main doors hardwood	each	2.00	KSh 42,000	KSh 84,000
Aluminium-framed sliding windows	m ²	33.00	KSh 9,500	KSh 313,500
			Trade subtotal	KSh 523,500

G — ELECTRICAL

Description	Unit	Qty	Rate	Amount
Mains intake, distribution board, MCB/RCD set, earthing	item	1.00	KSh 165,000	KSh 165,000
Switchgear & sub-mains for 16.8 kVA design demand	kVA	16.80	KSh 9,500	KSh 159,600
Lighting points & circuits (~1100 W LED design)	point	44.00	KSh 2,500	KSh 110,000
Socket outlets & ring final circuits (~5500 W)	point	40.00	KSh 2,800	KSh 112,000
Conduit, trunking, accessories & cable terminations	m ²	440.00	KSh 280	KSh 123,200
Lightning protection: air terminal, down conductor, ear	item	1.00	KSh 95,000	KSh 95,000
			Trade subtotal	KSh 764,800

H — PLUMBING

Description	Unit	Qty	Rate	Amount
Cold + hot water reticulation, copper/PEX, valves & fit	fixture-unit	10.00	KSh 24,000	KSh 240,000
Sanitary fixtures (WC, WHB, shower, sink, taps) — suppl	fixture-unit	10.00	KSh 18,500	KSh 185,000
Septic tank + soak-pit (4.8 m ³ + Ø8.24 m)	item	1.00	KSh 285,000	KSh 285,000
Stormwater drainage, gutters, downpipes & RWH connectio	m ²	231.00	KSh 480	KSh 110,880
Hot water system — solar/gas heater + pipework (KS 1893	each	2.00	KSh 65,000	KSh 130,000
			Trade subtotal	KSh 950,880

Cost summary

Item	Amount
A — PRELIMINARIES	KSh 1,440,000
B — SUBSTRUCTURE	KSh 4,663,120
C — SUPERSTRUCTURE	KSh 6,155,821
D — ROOFING	KSh 1,632,950
E — FINISHES	KSh 1,573,056
F — JOINERY	KSh 523,500
G — ELECTRICAL	KSh 764,800
H — PLUMBING	KSh 950,880
Subtotal of works	KSh 17,704,127
Contingency 7.5 %	KSh 1,327,810

Professional fees 8 % (Architect + Engineer + QS)	KSh 1,416,330
Sub-total before VAT	KSh 20,448,267
VAT 16 %	KSh 3,271,723
GRAND TOTAL	KSh 23,719,989
Cost per m ² (KES)	KSh 107,818 / m ²

6. STATUTORY COMPLIANCE PACK

All approvals required under Kenyan law before construction. The contractor and professional team shall ensure each item is closed prior to mobilisation.

Approval	Authority / Statute	Documents required
1. County Planning Approval	County Govt (Building Plans Section)	Stamped architectural & structural drawings, location plan, ownership docs, NCA
2. NCA Project Registration	National Construction Authority Act 2011	NCA-1 form, contractor registration certificate, levy 0.5 % of contract sum.
3. NEMA Project Report (PR)	EMCA 1999, EIA Regs 2003	Required for projects > 5 ha or near watercourses; PR submitted to NEMA county
4. Public Health Approval	Public Health Act Cap. 242	Septic system & water supply review by County Public Health Office.
5. EPRA / KPLC Energy Connection	EPRA Regulations	Electrical drawings & wayleave application; final inspection by EPRA-registered C
6. Fire Safety Certificate	County Fire Marshal	Means-of-escape, extinguisher schedule, hydrant layout (NFPA 241 / Kenya Fire
7. DOSHS Notification	OSHA 2007 §41	Notification before commencement; site safety plan, induction registers, ISO 4500

6.1 Professional certification (signature pages)

Role	Name & Reg. No.	Signature & Stamp	Date
Architect (BORAQS)	— NAME + REG. NO. REQUIRED	[Wet signature + official stamp]	Date: _____
Structural Engineer (EBK)	— NAME + REG. NO. REQUIRED	[Wet signature + official stamp]	Date: _____
Quantity Surveyor (BORAQS)	— NAME + REG. NO. REQUIRED	[Wet signature + official stamp]	Date: _____
Site Manager / Contractor (NCA)	— to be appointed before mobilisation	[Wet signature + official stamp]	Date: _____
Client / Developer	Smoke Test	[Signature]	Date: _____

■ ■ **Wet signatures & official practice stamps are mandatory.** Submission to County Government Building Plans Section requires each professional listed above to sign and stamp this page (and every drawing sheet) IN PERSON, in indelible ink, using the practising stamp issued by their licensing board. Digital signatures, typed names and image scans are NOT acceptable for statutory plan approval.

I certify that the design has been carried out in accordance with the relevant codes applicable to Kenya (including the Eurocodes / British Standards referenced in §2) and that I take professional responsibility for the section of work corresponding to my registration.

6.2 Required annexures (attach before submission)

The items below must be ATTACHED as certified copies to the submission pack. Tick each box once the document has been collected and verified by the project lead. Submissions with missing annexures are rejected at intake.

	Annexure	Notes
[]	Title deed / land ownership document	Certified copy from registry
[]	Survey / mutation drawing	Stamped by Licensed Land Surveyor
[]	Architectural drawings (stamped)	Each sheet stamped by BORAQS-registered architect
[]	Structural drawings & calculations	Stamped by EBK-registered engineer
[]	BOQ & cost plan	Stamped by BORAQS
[]	Soil investigation report (boreholes/SPT/lab)	Mandatory before foundation works
[]	Contractor registration certificate	NCA registration
[]	Builder's All-Risk insurance	Cover >= contract sum, valid for project duration
[]	Public-liability insurance	Cover >= KSh 10 M (or local equivalent)
[]	Workmen's compensation policy	Per applicable labour law
[]	Environmental Project Report (NEMA/EIA)	If site near watercourse / > 5 ha
[]	Fire-safety report & extinguisher schedule	Stamped by registered fire-safety consultant
[]	Public-health approval (septic / sewer)	Public Health Officer sign-off
[]	Energy / utility connection application	Power, water, telecoms wayleave
[]	Site safety plan & DOSHS notification	Per OSH Act 2007 §41 (or local equivalent)

7B. QUANTITY SURVEYOR QUOTATION — QS-20260427-1343

Quotation No.	QS-20260427-1343
Issued	2026-04-27
Validity	30 days from date of issue
From	EIMS Construction Solutions
To (Client)	Smoke Test
Project	Residential · 220 m² · 2 storey · 1 unit(s)
Location	Nakuru, Kenya, Other
Payment Terms	50/40/10
Estimated Duration	4 months
Currency	KES

7B.1 Scope of Work

- Site clearing and preparation
- Foundation and substructure works
- Superstructure: 2-storey residential building (220m²)
- Roofing and waterproofing
- Electrical installation and wiring
- Plumbing, drainage and water supply
- Internal and external finishes
- Painting and decorating
- External works: paving, landscaping, fencing
- HVAC and mechanical systems
- Testing, commissioning and handover

7B.2 Bill of Quantities — Trade Sections

Note: 1 line item(s) priced at QS-overridden rates (marked ★ below).

A. SUBSTRUCTURE & SUPERSTRUCTURE

#	Description	Unit	Qty	Rate	Amount
1	Portland Cement (50kg)	bag	1,045.80	KES1,308	KES1,367,582
2	Reinforcement Steel Y12 (12mm)	kg	5,809.51	KES135	KES786,491
3	Reinforcement Steel Y16 (16mm)	kg	5,083.32	KES138	KES703,836
4	Reinforcement Steel Y10 (10mm)	kg	3,630.95	KES131	KES474,819
5	Binding Wire (gauge 18)	kg	224.28	KES231	KES51,757
6	BRC Mesh A142 (4.8m×2.4m)	sheet	11.55	KES4,308	KES49,754
7	Fine River Sand	m³	70.51	KES3,846	KES271,192
8	Coarse Aggregate (20mm)	m³	141.02	KES4,923	KES694,253
9	Concrete Blocks 200mm (8")	piece	2,935.80	KES154	KES451,673
10	Concrete Blocks 100mm (4")	piece	3,523.80	KES100	KES352,380
11	Formwork Plywood 18mm	sheet	146.05	KES3,385	KES494,324
12	Timber 50×100mm (Softwood)	m	1,117.60	KES431	KES481,429
13	DPC Polythene (1000 gauge)	m²	26.46	KES185	KES4,885

#	Description	Unit	Qty	Rate	Amount
14	Hardcore / Murram	m ³	24.20	KES2,769	KES67,015
15	Anti-termite Chemical	litre	57.75	KES1,846	KES106,615
16	Waterproof Membrane	m ²	46.20	KES1,308	KES60,415
17	Precast Concrete Lintel (1200mm)	piece	13.26	KES1,231	KES16,320
				Subtotal	KES6,434,741

B. ELECTRICAL INSTALLATION

#	Description	Unit	Qty	Rate	Amount
18	Copper Wire 2.5mm ² (single core)	m	173.04	KES100	KES17,304
19	Copper Wire 1.5mm ² (single core)	m	104.03	KES69	KES7,202
20	PVC Conduit 20mm	m	140.70	KES77	KES10,823
21	Switch 1-Gang 1-Way	piece	12.12	KES538	KES6,526
22	Socket Outlet 13A Double	piece	12.12	KES1,000	KES12,120
23	LED Downlight 9W	piece	12.24	KES1,231	KES15,065
24	Circuit Breaker MCB 20A	piece	4.04	KES1,000	KES4,040
25	RCCB/ELCB 40A 30mA	piece	1.01	KES3,846	KES3,885
26	Distribution Board 12-Way	piece	1.01	KES10,000	KES10,100
27	Earth Rod (1.5m Copper)	piece	1.01	KES2,308	KES2,331
28	Smoke Detector	piece	6.06	KES2,769	KES16,782
				Subtotal	KES106,176

C. PLUMBING & DRAINAGE

#	Description	Unit	Qty	Rate	Amount
29	PPR Pipe 20mm (Hot/Cold)	m	39.90	KES277	KES11,049
30	PVC Pipe 110mm (4m)	length	6.30	KES1,846	KES11,631
31	PVC Pipe 50mm (4m)	length	3.15	KES846	KES2,665
32	Toilet (Close-Coupled)	set	2.02	KES13,077	KES26,415
33	Washbasin (Pedestal)	piece	2.04	KES8,462	KES17,262
34	Shower Mixer Valve	piece	2.02	KES6,923	KES13,985
35	Shower Head (Rain 200mm)	piece	2.02	KES5,385	KES10,877
36	Basin Mixer Tap	piece	2.02	KES4,615	KES9,323
37	Kitchen Sink (SS Double Bowl)	piece	1.01	KES10,000	KES10,100
38	Kitchen Mixer Tap	piece	1.01	KES6,154	KES6,215
39	Water Tank (Polyethylene 2000L)	piece	1.01	KES30,769	KES31,077
40	Water Heater (50L Electric)	piece	1.01	KES23,077	KES23,308
41	Floor Drain (SS 150mm)	piece	5.10	KES923	KES4,708
42	Inspection Chamber Cover (450mm)	piece	2.02	KES3,846	KES7,769
				Subtotal	KES186,384

D. FINISHES & JOINERY

#	Description	Unit	Qty	Rate	Amount
43	Ceramic Floor Tile 600×600mm	m ²	242.00	KES2,769	KES670,154
44	Ceramic Wall Tile 300×600mm	m ²	35.20	KES2,308	KES81,231
45	Tile Adhesive (20kg)	bag	52.50	KES923	KES48,462
46	Tile Grout (5kg)	bag	32.55	KES615	KES20,031
47	Interior Emulsion Paint (20L)	can	4.40	KES5,385	KES23,692
48	Exterior Weather Paint (20L)	can	3.30	KES6,923	KES22,846
49	Primer/Undercoat (20L)	can	3.30	KES4,308	KES14,215
50	Cement Plaster (Wall 15mm)	m ²	389.28	KES692	KES269,502
51	Gypsum Board Ceiling (9.5mm)	m ²	237.60	KES1,846	KES438,645
52	Flush Door Leaf (900×2100)	piece	7.14	KES6,923	KES49,431
53	Timber Door Frame (900×2100)	piece	7.14	KES5,385	KES38,446
54	Security Steel Door	piece	1.01	KES18,462	KES18,646
55	Door Lock Set (Mortise)	piece	7.07	KES2,769	KES19,578
56	Door Hinges (3" Stainless, pair)	pair	14.14	KES615	KES8,701
57	Aluminum Window Frame	m ²	8.90	KES8,462	KES75,308
58	Float Glass 4mm (Clear)	m ²	9.07	KES1,231	KES11,163
59	Putty / Wall Filler (5kg)	bag	9.45	KES769	KES7,269
60	Granite Countertop (25mm)	m ²	3.15	KES12,308	KES38,769
				Subtotal	KES1,856,091

E. EXTERNAL WORKS & ROOFING

#	Description	Unit	Qty	Rate	Amount
61	Roofing Sheet (Box Profile 3m)	sheet	48.30	KES2,308	KES111,461
62	Ridge Cap	piece	11.55	KES769	KES8,885
63	Guttering (PVC 100mm, 3m)	length	7.35	KES1,231	KES9,046
64	Downpipe (PVC 75mm, 3m)	length	4.20	KES923	KES3,877
65	Fascia Board (225mm Timber)	m	44.10	KES615	KES27,138
66	Concrete Paving Block (60mm)	m ²	34.65	KES2,308	KES79,961
67	Topsoil	m ³	17.32	KES3,846	KES66,615
68	Grass Sod / Turf	m ²	60.50	KES615	KES37,230
69	Perimeter Wall Block (200mm)	piece	2,201.85	KES185	KES406,506
70	Septic Tank (Prefab 3000L)	piece	1.01	KES76,923	KES77,692
71	Soakaway Pit (Precast Rings)	set	1.01	KES30,769	KES31,077
72	Manhole Cover (Heavy Duty)	piece	2.02	KES4,615	KES9,323
				Subtotal	KES868,813

F. MECHANICAL & HVAC

#	Description	Unit	Qty	Rate	Amount
73	Split AC Unit (12000 BTU)	set	4.04	KES53,846	KES217,538
74	Ceiling Fan (1200mm)	piece	6.06	KES5,385	KES32,631
75	Exhaust Fan (200mm)	piece	3.03	KES3,077	KES9,323

#	Description	Unit	Qty	Rate	Amount
76	Fire Extinguisher (CO2 5kg)	piece	1.01	KES6,923	KES6,992
77	Water Pump (0.75kW)	piece	1.01	KES18,462	KES18,646
				Subtotal	KES285,131

7B.3 Cost Summary

Material Cost	KES9,737,348
Labour Cost	KES2,190,903
Equipment	KES778,988
Overhead	KES1,431,391
Contingency	KES1,413,863
Subtotal	KES15,552,492
Contractor Profit	KES3,110,498
Gross Total	KES18,662,991
GRAND TOTAL	KES18,662,991
Cost per m ² (KES)	KES84,832 / m ²

7B.4 Payment Schedule

Stage	%	Amount
Mobilization & Foundation	50%	KES9,331,495
Superstructure & Roofing	40%	KES7,465,196
Completion & Handover	10%	KES1,866,299

7B.5 Exclusions

- Land acquisition and survey costs
- Government permit and approval fees
- Furniture, fittings and equipment (FF&E;)
- Temporary power and water supply during construction
- Professional fees (architect, engineer, QS)
- Land rates, taxes and insurance

7B.6 Terms & Conditions

- Prices are valid for the stated validity period only
- Variations to scope will be priced separately
- Payment to be made within 14 days of each stage certificate
- Retention of 5% to be released 6 months after practical completion
- Contractor to provide 12-month defects liability period
- All works to comply with local building regulations
- Client to provide clear site access and water/power connection
- All prices shown in KES (Kenyan Shilling)

For and on behalf of:

Accepted by client:

EIMS Construction Solutions

Signature: _____

Name: _____

Title: Quantity Surveyor

Date: _____

Smoke Test

Signature: _____

Name: _____

Title: _____

Date: _____

8. PROJECT SUMMARY

Phase phase_1: Site Analysis

water_table_depth	3.5
water_table_source	regional_estimate
water_table_note	Regional estimate from climate/soil model -- verify with on-site geotechnical su
soil_type	clay
vegetation_coverage	0.65
slope_angle	8.25
suitability_pct	90
nearby_structures	survey_required
land_use	residential
climate_zone	tropical
analysis_method	regional_data_model
data_source	EIMS Regional Database + GPS Coordinates

Phase phase_2: Geotechnical Design

bearing_capacity_kPa	150
soil_classification	CH
recommended_foundation	STRIP
settlement_mm	40
slope_stability	stable
water_table_depth	3.5
excavation_depth	2.8

Phase phase_3: Foundation Design

foundation_type	STRIP + GROUND BEAM
foundation_depth_m	1.3
footing_width_m	1.82
total_building_load_kN	1650.0
bearing_capacity_kPa	150
safety_factor	3.0
concrete_volume_m3	42.9
reinforcement_steel_kg	3861.0
soil_type	clay
water_table_factor	standard
design_standard	BS 8004 / Eurocode 7
bim_ready	True

Phase phase_4: Floor Plans

floor_count	2
total_area_sqm	220.0
bathrooms	3
rooms	{'bedrooms': 4, 'bathrooms': 3, 'living_areas': 1, 'kitchen': 1, 'extra_rooms':

style	modern
area_per_unit	220.0
building_code_compliance	True
area_efficiency_percent	80.0
cost_per_sqm_estimate	calculated_in_costing_phase

Phase phase_5: Electrical Design

design_standard	IEC 60364 / KS IEC 60364
connected_load_kW	24.6
diversity_factor	0.65
design_demand_kW	15.99
design_demand_kVA	16.8
main_breaker_A	75
power_factor	0.95
occupants	8
lighting_W	1100.0
sockets_W	5500.0
cooker_W	6000
water_heater_W	6000

Phase phase_6: Plumbing & Drainage Design

design_standard	BS 8233 occupancy + BS EN 12056 drainage
occupants	8
daily_water_L	1600
septic_volume_m3	4.8
septic_dim	2.5 × 1.2 × 1.6 m (L×W×D)
soak_pit_area_m2	53.33
soak_pit_diameter_m	8.24
fixture_units	{'Water Closet (WC)': 2, 'Wash-Hand Basin': 2, 'Shower / Bath': 2, 'Kitchen Sink
stormwater_note	Roof catchment 220 m² — gutter sized per Kenya NCC (rainfall 50 mm/h design inte
note	Single canonical plumbing design — supersedes legacy MEPDesigner output.

Phase phase_7: BOQ — KQS-2025

currency	KES
subtotal	15175226.98
contingency_7_5pct	1138142.02
professional_fees_8pct	1214018.16
sub_before_vat	17527387.16
vat_16pct	2804381.95
grand_total	20331769.11
cost_per_m2	92417.13
sections_count	8
note	BOQ \$5 is the single source of truth for cost. Phase 12 mirrors this total.

Phase phase_8: Infrastructure

solar_system	{'panels_kw': 4.4, 'panels_count': 11, 'battery_kWh': 6.6, 'estimated_cost_usd':
borehole_system	{'depth_meters': 8, 'estimated_yield_lpm': 15, 'pump_kw': 3, 'estimated_cost_usd
grid_option	{'connection_cost_usd': 260, 'monthly_tariff_estimate_usd': 33}
recommendation	solar_grid_hybrid
estimated_annual_saving_usd	819.84

Phase phase_9: Landscape Design

lawn_area_m2	77.0
hardscape_area_m2	22.0
garden_area_m2	33.0
plants_count	264
irrigation_system	True
recommended_style	tropical_lush
estimated_cost_usd	759
maintenance_monthly_usd	15

Phase phase_10: Permits & Compliance

jurisdiction	Kenya
building_code	Kenya Building Code 2020
building_code_compliance	REQUIRES_VERIFICATION
permits_required	6
estimated_timeline_days	45
estimated_fees_usd	352
setback_compliance	{'front_m': 5, 'side_m': 2, 'rear_m': 3}
required_documents	['Architectural drawings (stamped)', 'Structural calculations (engineer-certifie

Phase phase_11: 3D Visualization

status	ready
model_endpoint	/api/drawings/3d-model

Phase phase_12: Live Costing — reconciled to BOQ

currency	KES
symbol	KSh
grand_total	20331769.11
cost_per_m2	92417.13
subtotal	15175226.98
contingency_7_5pct	1138142.02
professional_fees_8pct	1214018.16
sub_before_vat	17527387.16
vat_16pct	2804381.95
reconciled_with	Section 5 BOQ (canonical)
note	Phase 12 is mathematically identical to BOQ \$5 grand total — no method discrepan

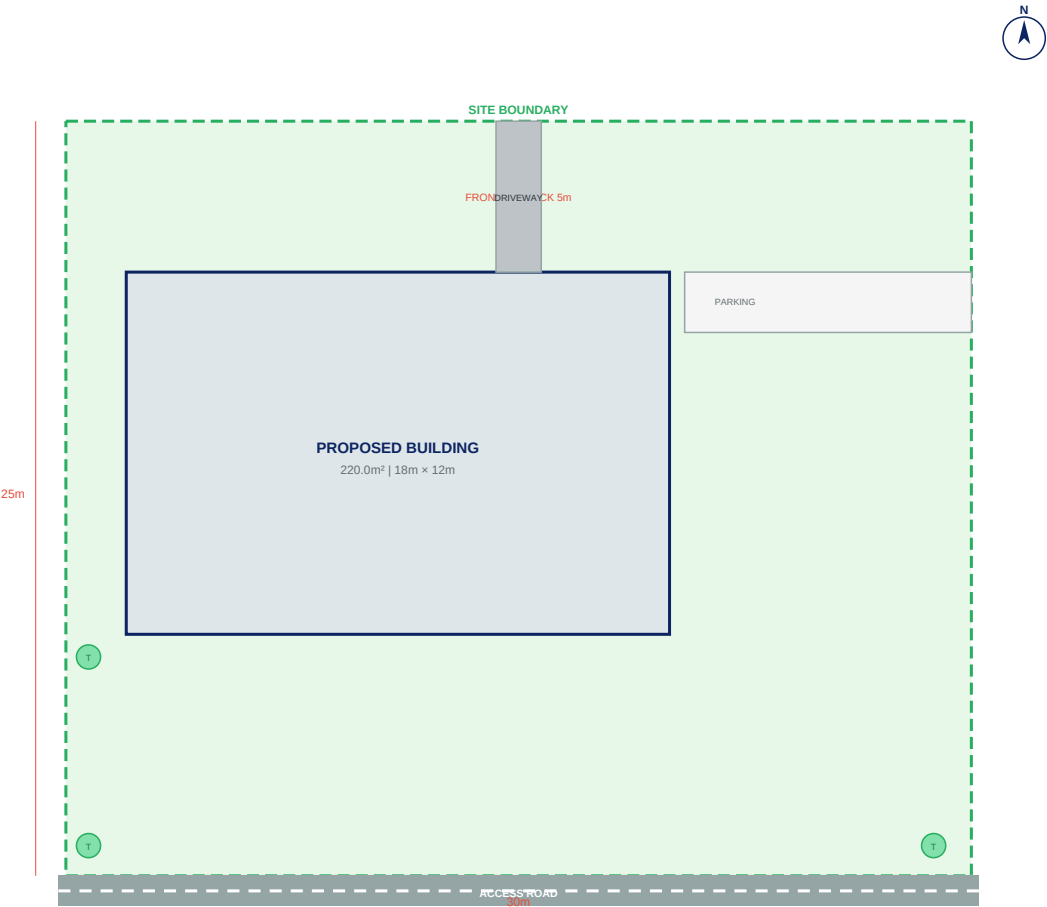
Phase phase_13: Integration Layer

status	complete
export_formats	['IFC2x3', 'FBX', 'DXF', 'PDF', 'XLSX']

9. SITE PLAN

Site layout showing building footprint, access roads, setbacks and landscaping. Area: 220 m²

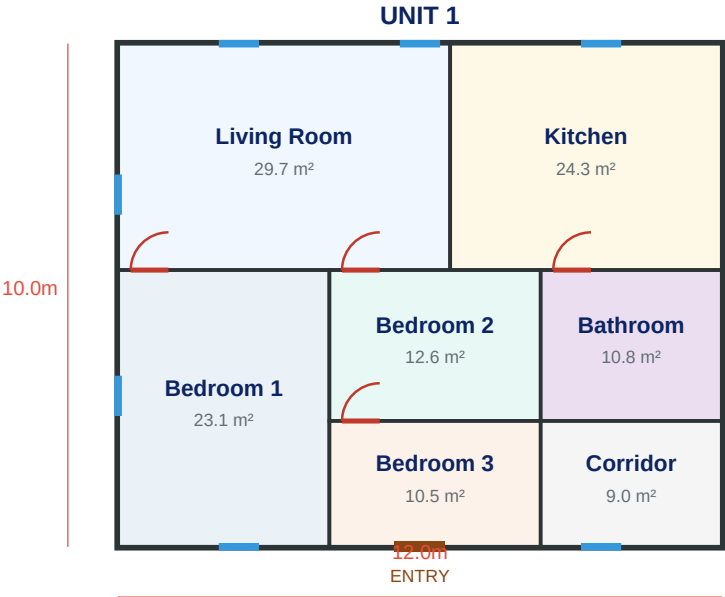
SITE PLAN - Nakuru, Kenya | Plot: 30m×25m



10. ARCHITECTURAL FLOOR PLAN

Ground floor layout — 1 unit(s), 4 bedrooms, showing room dimensions, doors, windows and circulation.

FLOOR PLAN - 220.0m² | 1 Units | 2 Storey(s)



EMERSON EIMS BUILDING SUITE PRO

Area: 220.0m² | 1 Units × 4 Bed

Floor Plan | Scale 1:50 | 2026-04-27 | Drawing: FP-001

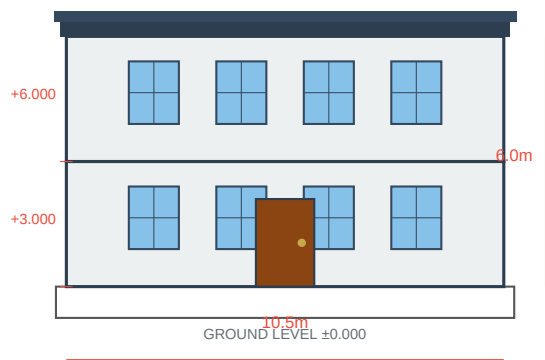
IFC Ready | BIM Level 2

11. BUILDING ELEVATIONS

4a. NORTH ELEVATION

Front view showing 2 storey(s), window/door openings, roof profile and floor levels.

NORTH ELEVATION - 2 Stories | Style: Modern



EMERSON EIMS | NORTH ELEVATION

Scale 1:50 | 2026-04-27 | Drawing: EL-N01

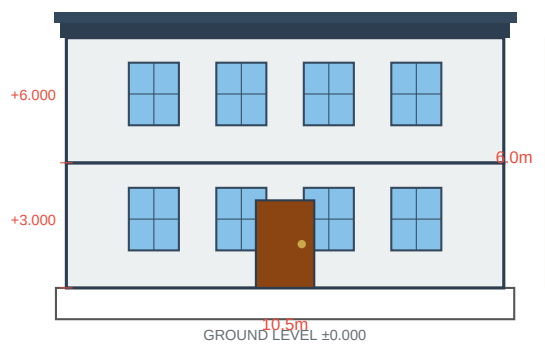
Stories: 2 | Height: 6.0m

Style: Modern | IFC Ready

4b. EAST ELEVATION

Side view showing building depth, roof slope and storey heights.

EAST ELEVATION - 2 Stories | Style: Modern



EMERSON EIMS | EAST ELEVATION

Scale 1:50 | 2026-04-27 | Drawing: EL-E01

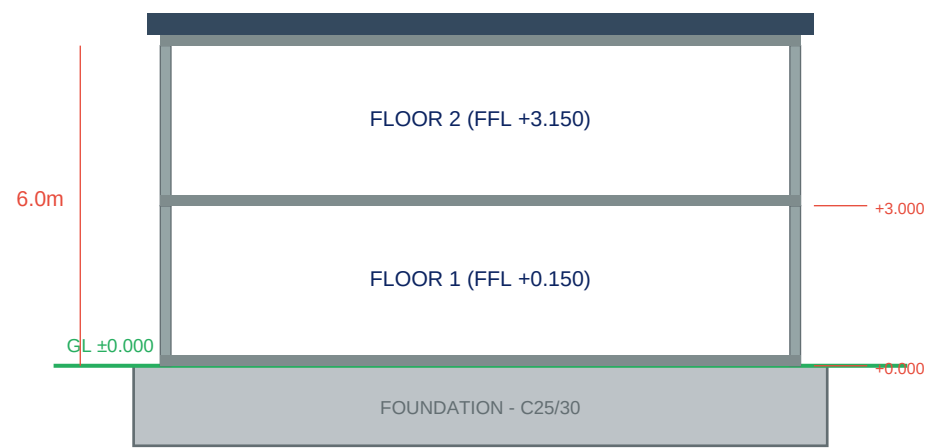
Stories: 2 | Height: 6.0m

Style: Modern | IFC Ready

12. BUILDING CROSS SECTION

Vertical cut through building showing floor slabs, storey heights (2 floors), foundation depth, roof structure and structural members.

CROSS SECTION A-A | 2 Stories



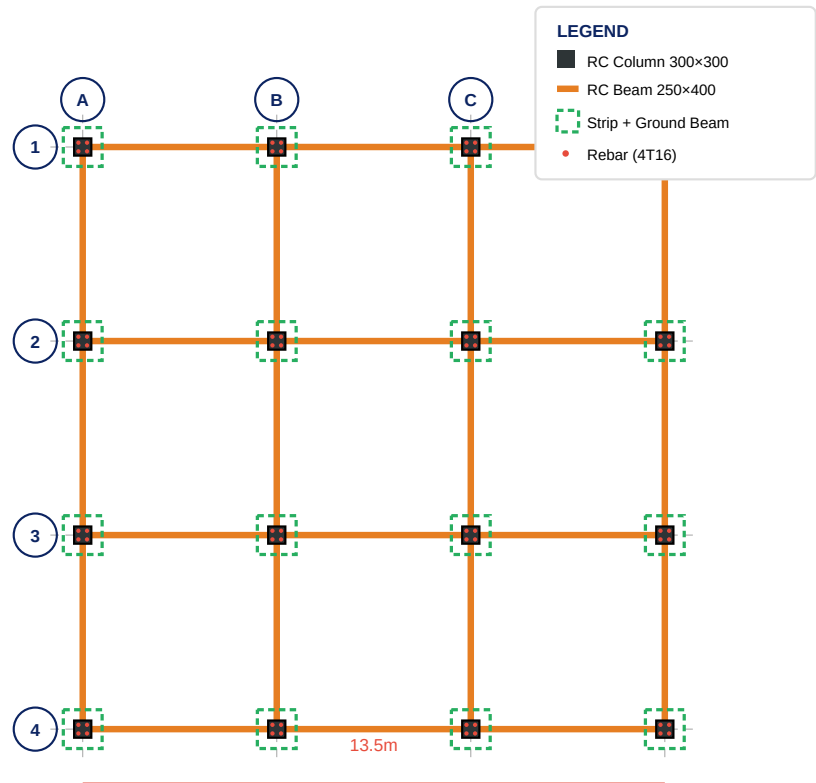
EMERSON EIMS | BUILDING SECTION A-A

2026-04-27 | SEC-001 | Scale 1:50

13. STRUCTURAL LAYOUT

Column grid, beam layout and foundation pads. Foundation type: Strip + Ground Beam. All dimensions in metres. Concrete grade C25/30, Steel Y16.

STRUCTURAL LAYOUT - Strip + Ground Beam Foundation | 2F | Grid 4.5m×4.5m



EMERSON EIMS | STRUCTURAL LAYOUT

Grid: 4.5m × 4.5m | Columns: 16 | 2026-04-27 | STR-001

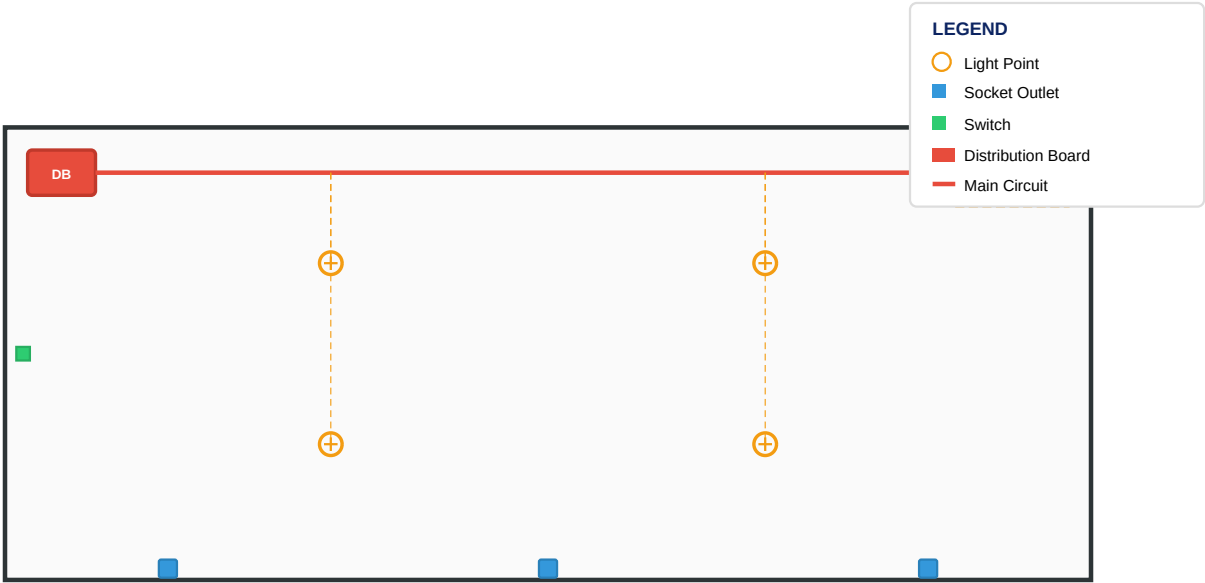
Concrete: C25/30 | Steel: Y16

Foundation: Strip + Ground Beam | FOS ≥ 3.0

14. MEP LAYOUT — ELECTRICAL

Electrical distribution layout showing DB positions, circuit runs, socket/light points and earthing.

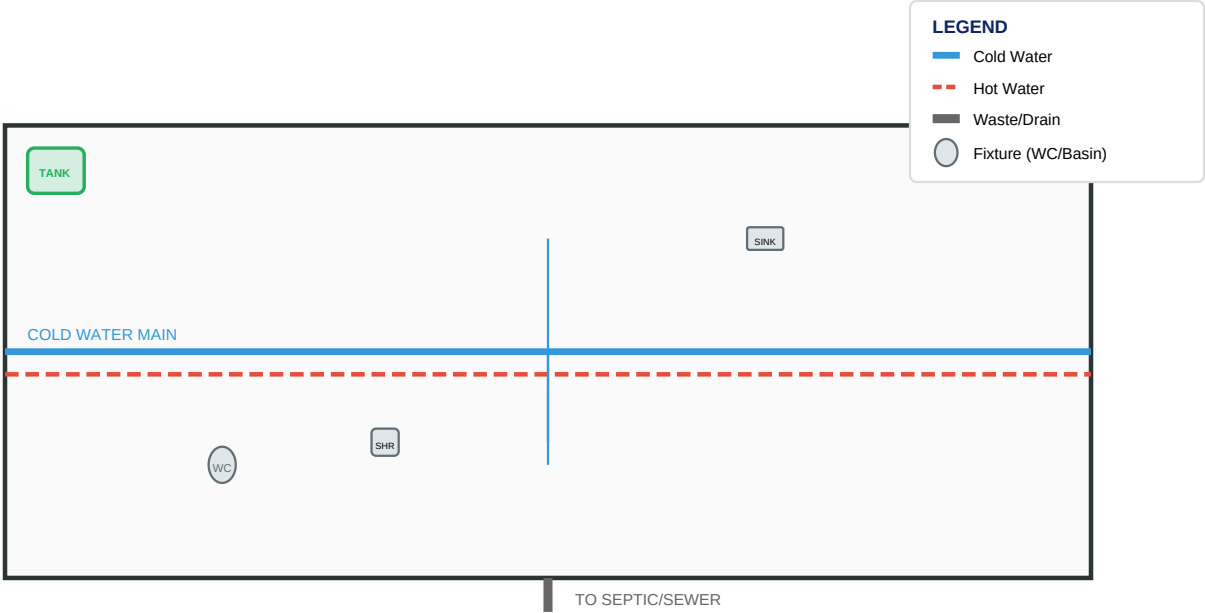
ELECTRICAL LAYOUT - 220.0m² | 1 Units



15. MEP LAYOUT — PLUMBING

Plumbing layout showing supply/drain runs, fixture locations, water heater and main shutoff valve.

PLUMBING LAYOUT - 220.0m² | 1 Units



EMERSON EIMS | PLUMBING LAYOUT

Scale 1:50 | 2026-04-27 | PLUMBING-001

16. COST SUMMARY

Item	Amount (KES)
Grand Total	KSh 20,331,769.11
Cost Per M2	KSh 92,417.13
Subtotal	KSh 15,175,226.98
Contingency 7 5Pct	KSh 1,138,142.02
Professional Fees 8Pct	KSh 1,214,018.16
Sub Before Vat	KSh 17,527,387.16
Vat 16Pct	KSh 2,804,381.95

This report was generated by EMERSON EIMS Building Suite Pro. All drawings are indicative and must be verified by a licensed engineer before construction.